OpenH2O

AI ENVIRONMENTAL IMPACT TRACKER

Claude Code Build Specification · Tech Awareness Association

v1.0 · March 2026

1. Project Overview

OpenH2O is a standalone, client-side web application that allows any Claude user to upload their exported Claude conversation history (JSON) and receive a detailed personal environmental impact report — showing estimated carbon emissions, water usage, and energy consumption from their AI usage over time.

Project Name	OpenH2O — AI Environmental Impact Tracker
Owner	Tech Awareness Association (TAA) — Shrewsbury, MA
Tech Stack	React (Vite) + Tailwind CSS + React Router
Data Processing	100% client-side — no server, no data storage, no uploads to any backend
Hosting	Static deployment (Vercel)
Audience	General Claude users (anyone, any age, any technical level)
Monetization	Free, nonprofit — TAA branded, open source on GitHub

Core user flow:

- User visits OpenH2O web app
- User exports their Claude data from claude.ai (Settings → Privacy → Export Data)
- User uploads the conversations.json file to OpenH2O
- App parses conversations client-side, tokenizes message text, applies energy/carbon/water coefficients
- App displays a rich personal impact dashboard with relatable real-world comparisons
- User can explore breakdown by conversation, date range, and message type

2. UI Design System

OpenH2O inherits the exact visual design language from the Susu app (TAA's senior tech literacy platform). This ensures brand consistency across all TAA tools.

2.1 Color Palette

token name	hex value	usage
`sunshine`	`#FAC206`	Primary CTA buttons, accents, active nav state, key metric highlights
`sky`	`#16C0FF`	Secondary accents, progress bars, info callouts, water metric color
`coral`	`#FB4B5F`	Warning states, CO2 metric color, destructive actions
`navy`	`#2C3E50`	Sidebar background, headings, borders, primary text on light backgrounds
`cream`	`#F8F3F1`	Page background, card alternates
`ink`	`#181818`	Body text, primary dark text
`slate`	`#8B9B9C`	Secondary text, labels, metadata
`green (eco)`	`#2ECC71`	Energy metric color, positive/low-impact states

2.2 Typography

- Font family: Inter (Google Fonts), fallback: system-ui, sans-serif
- Headings: font-black (900 weight), uppercase, tight tracking
- Body: font-bold (700) for labels/metadata, normal weight for body copy
- Monospace: Courier New for file names, token counts, coefficient values

2.3 Layout & Component Style (Boxy Aesthetic)

All components must use the Susu boxy design language:

- border-4 border-navy on all cards and containers
- Shadow offsets: shadow-[8px_8px_0px_0px_#FAC206] for featured cards, shadow-[4px_4px_0px_0px_#2C3E50] for standard
- No border-radius anywhere — all elements are sharp-cornered rectangles
- Buttons: border-4 border-navy, font-black, uppercase, tracking-wider, min-h-12
- Active/selected states: bg-sunshine text-ink with navy border
- Hover states: shadow offset appears or increases, slight -translate-y-0.5
- Sidebar: bg-navy with sunshine active indicator on left border

2.4 Layout Structure

Mirror Susu's exact Layout.jsx structure:

- Desktop: fixed left sidebar (w-60) + main content area (ml-60)

- Mobile: top bar + bottom nav (4 items max)
- Page background: bg-cream (#F8F3F1)
- Content max-width: max-w-5xl centered with px-4 md:px-8 lg:px-12 py-8
- Sidebar footer: TAA credit with link to techawarenessma.com

2.5 OpenH2O-Specific Metric Color Coding

Each environmental metric has a dedicated color used consistently across all charts, cards, and labels:

- Carbon (CO2e): coral #FB4B5F — matches danger/warning semantics
- Water (liters): sky #16C0FF — intuitive water association
- Energy (kWh): green #2ECC71 — standard energy/sustainability color

3. Data Pipeline & Calculation Engine

3.1 Claude Export JSON Schema

Based on analysis of actual Claude data exports, the conversations.json has this structure:

```
conversations.json  →  Array of conversation objects

conversation {
  uuid: string
  name: string                    // conversation title
  summary: string                 // AI-generated summary
  created_at: ISO timestamp
  updated_at: ISO timestamp
  account: { uuid: string }
  chat_messages: [
    {
      uuid: string
      content: [ { type, text, ... } ]
      sender: 'human' | 'assistant'
      created_at: ISO timestamp
      updated_at: ISO timestamp
      attachments: []
      files: []
    }
  ]
}
```

Critical notes:

- No token counts are included in the export — tokenization must be done client-side
- No model name per conversation is included — Claude Sonnet 4.6 coefficients are used as the default with a disclosure
- message.content is an array of content blocks; extract text from blocks where type === 'text'
- sender values are 'human' (user prompts) and 'assistant' (Claude responses)

3.2 Tokenization

Client-side tokenization uses the gpt-tokenizer npm package (GPT-4 tiktoken BPE encoding). Claude's tokenizer is not publicly available, but GPT-4 tiktoken produces token counts within ~5% for typical English text — acceptable for estimation purposes. This approximation is disclosed to users.

- Install: npm install gpt-tokenizer
- Usage: import { encode } from 'gpt-tokenizer'; const tokens = encode(text).length
- Count input_tokens from sender === 'human' messages
- Count output_tokens from sender === 'assistant' messages
- Store per-conversation and aggregate totals

3.3 Energy & Environmental Coefficients

All coefficients are stored in src/data/coefficients.js as a versioned, documented object. This file is designed for easy future updates by maintainers — no other files need to change when research is updated.

Variable	Value	Source / Notes
energy_per_output_token_wh	0.0000030 Wh	Luccioni et al. 2023; output tokens ~3-5x input cost — use output as primary driver
energy_per_input_token_wh	0.0000010 Wh	Luccioni et al. 2023; input pass is single forward pass, significantly cheaper
pue_multiplier	1.2	Anthropic datacenter PUE estimate (industry average 1.1–1.5); applied to raw compute
water_per_kwh_liters	1.9	Direct cooling: 1,900 L/MWh (EESI 2024); converted to per-kWh
carbon_per_kwh_gco2e	350	US average grid intensity gCO2e/kWh (EPA eGRID 2023); Anthropic uses US East
model_label	Claude Sonnet 4.6	Default model; no per-message model data in exports
coefficients_version	1.0	Increment when updating any value; displayed in app UI
last_updated	March 2026	Date coefficients were last reviewed against literature

3.4 Calculation Formulas

Core calculation chain (executed per conversation, aggregated for totals):

```
// Step 1: Tokenize
inputTokens  = encode(humanMessages.join(' ')).length
outputTokens = encode(assistantMessages.join(' ')).length

// Step 2: Raw energy
rawEnergy_wh = (inputTokens  * energy_per_input_token_wh)
             + (outputTokens * energy_per_output_token_wh)

// Step 3: Apply PUE
```

```
totalEnergy_wh  = rawEnergy_wh * pue_multiplier
totalEnergy_kwh = totalEnergy_wh / 1000

// Step 4: Water & Carbon
water_liters  = totalEnergy_kwh * water_per_kwh_liters
carbon_gco2e  = totalEnergy_kwh * carbon_per_kwh_gco2e
```

3.5 Real-World Comparisons

All metrics are translated into relatable units. These comparisons are stored in src/data/comparisons.js:

Metric	Comparison Unit	Formula / Source
Carbon (gCO2e)	Miles driven (gas car)	÷ 404 gCO2e/mile (EPA 2023)
Carbon (gCO2e)	Smartphone charges	÷ 8.22 gCO2e/charge (avg 12Wh × 685g/kWh)
Water (liters)	Water bottles (500mL)	÷ 0.5
Water (liters)	Minutes of shower	÷ 8 liters/min (avg shower)
Energy (kWh)	LED bulb hours	÷ 0.01 kWh/hr (10W LED)
Energy (kWh)	Hours of laptop use	÷ 0.05 kWh/hr (avg laptop)

4. Pages & Routes

Route	Page Name	Purpose
`/`	**Landing / Upload**	Hero page explaining OpenH2O + file upload CTA
`/processing`	**Processing**	Progress indicator while JSON is parsed and tokenized
`/dashboard`	**My Impact**	Main dashboard: total metrics, charts, timeline
`/breakdown`	**Conversation List**	Per-conversation table with sortable columns
`/insights`	**Insights**	Comparisons, trends, efficiency tips
`/methodology`	**Methodology**	Full disclosure of coefficients, assumptions, sources
`/about`	**About**	TAA mission, OpenH2O purpose, GitHub link

5. Page Specifications

5.1 / Landing & Upload Page

Purpose: First impression + onboarding. Explains the tool, builds trust, guides the user to export their data and upload it.

Hero Section (mirrors Susu landing — navy left panel + cream right)

> Left panel (navy background): OpenH2O logo + tagline + 3 feature bullets (Free · Private · No Account Needed)
>
> Right panel (cream background): Step-by-step upload guide (3 steps with numbered boxes)
>
> Step 1: Export your Claude data (with link to claude.ai settings screenshot guide)
>
> Step 2: Find conversations.json in the downloaded ZIP
>
> Step 3: Upload it here — results in seconds

Upload Zone

> Large bordered drop zone: border-4 border-dashed border-navy bg-white
>
> Drag-and-drop + click-to-browse support
>
> Accept only .json files
>
> On hover: bg-cream with sunshine border
>
> On file selected: show filename + file size + green checkmark
>
> CTA button: 'Analyze My Impact →' — bg-sunshine border-4 border-navy font-black uppercase
>
> Privacy notice below: 'Your data never leaves your device. Processing happens entirely in your browser.'

What to Expect Section (below upload)

> 3 metric preview cards showing what will be calculated: 🌡️ Carbon · 💧 Water · ⚡ Energy
>
> Each card: bg-white border-4 border-navy, metric icon, name, brief description
>
> Sample output teaser (blurred/mock dashboard screenshot or illustration)

5.2 /processing Processing Page

Purpose: Shown while the JSON is parsed and tokenized. Prevents user confusion during the ~1-5 second processing window.

- Full-page centered layout on cream background
- Large animated progress bar (sky color, border-4 border-navy)
- Status messages cycling through: 'Reading conversations...' → 'Counting tokens...' → 'Calculating your impact...' → 'Almost done!'
- Stats appearing as they compute: X conversations found, X messages, X total tokens
- No user action required — auto-redirects to /dashboard on completion

5.3 /dashboard My Impact Dashboard

The core page. Shows the user's total environmental impact across their entire Claude history.

Header Row

> Username (from users.json if available, otherwise 'Your Impact')
>
> Date range of conversations (earliest to latest from export)

> Total conversation count and total message count as subtitle metadata

Big 3 Metric Cards (top of page, 3-column grid)

> Card 1 — Carbon: large gCO2e number, coral accent, equivalent comparison (e.g., '= X miles driven')
>
> Card 2 — Water: large liters number, sky accent, equivalent comparison (e.g., '= X water bottles')
>
> Card 3 — Energy: large kWh number, green accent, equivalent comparison (e.g., '= X LED bulb hours')
>
> Each card: bg-white border-4 border-navy shadow-[8px_8px_0px_0px_ACCENT_COLOR]
>
> Font size for numbers: text-6xl md:text-7xl font-black
>
> Comparison text below in smaller font with emoji icon

Monthly Usage Timeline Chart

> Bar chart using recharts BarChart component
>
> X-axis: months, Y-axis: conversations per month
>
> Bars colored sunshine (#FAC206) with navy border stroke
>
> Clicking a bar filters the breakdown page to that month
>
> Chart container: bg-white border-4 border-navy p-6

Token Distribution Panel

> Two stat boxes side by side: Input Tokens (human messages) vs Output Tokens (Claude responses)
>
> Shows total count + % of total + energy contribution
>
> Visual bar showing input vs output ratio
>
> Note: Output tokens cost ~3x more energy than input — clearly labeled

Fun Facts Strip

> Horizontal scrolling or grid of 3-4 comparison callout boxes
>
> Examples: 'Your Claude usage this year = leaving a lightbulb on for X hours'
>
> 'That's the same water as X cups of coffee'
>
> bg-sunshine border-4 border-navy text-ink font-black — eye-catching, shareable

5.4 /breakdown Conversation Breakdown

Per-conversation table showing individual impact of each conversation.

- Searchable, sortable table: columns = Date, Title (truncated), Messages, Tokens, Energy (Wh), Water (mL), Carbon (gCO2e)
- Sort by any column — default sort: most recent first
- Search bar filters by conversation title
- Each row: clicking expands to show message count breakdown (input vs output tokens)
- Top 5 most impactful conversations highlighted with sunshine left border

- Pagination: 20 conversations per page

5.5 /insights Insights Page

Educational content tailored to the user's actual data.

- 'Your Usage Patterns' section: busiest days/times, avg conversation length, avg tokens per conversation
- 'How You Compare' section: compare user's per-conversation average against published research averages
- 'Reduce Your Impact' tips: 5 actionable tips with the user's data woven in ('You average X output tokens per conversation — asking for shorter answers could cut that by ~30%')
- 'The Bigger Picture' section: scale context — individual vs. systemic impact discussion, link to Methodology

5.6 /methodology Methodology Page

Full transparency page. Critical for TAA's educational mission and for tool credibility.

- Opening statement: 'These are estimates, not measurements. We use the best available published research and state all assumptions clearly.'
- Coefficients table: shows every coefficient, its value, source, and date
- Tokenization section: explains gpt-tokenizer approximation and its ~5% error range
- Formula walkthrough: step-by-step calculation with example numbers
- Uncertainty section: what we don't know (Anthropic's actual PUE, model specifics, caching effects)
- References section: full citation list (Luccioni et al., EESI, EPA eGRID, de Vries 2023)
- Version number and last-updated date prominently displayed

6. Component Architecture

6.1 Layout Component

Mirror Susu's Layout.jsx exactly, with OpenH2O-specific nav items:

- Sidebar nav items: Upload (home icon), My Impact (chart icon), Breakdown (list icon), Insights (lightbulb icon), Methodology (book icon), About (info icon)
- Sidebar shows upload status: if no data loaded yet, show 'No data uploaded' badge
- If data is loaded: show mini-stats in sidebar footer: 'X conversations · loaded MM/DD'

Component	File	Description
Layout	`components/Layout.jsx`	Sidebar + mobile nav shell (mirrors Susu Layout)
MetricCard	`components/MetricCard.jsx`	Big 3 impact metric display cards
UploadZone	`components/UploadZone.jsx`	Drag-and-drop JSON file upload
UsageChart	`components/UsageChart.jsx`	Monthly usage bar chart (recharts)
ConvoTable	`components/ConvoTable.jsx`	Sortable/searchable conversation breakdown

ComparisonBadge	`components/ComparisonBadge.jsx`	Real-world comparison callout boxes
ProgressBar	`components/ProgressBar.jsx`	Reused from Susu — unchanged
MethodologyRef	`components/MethodologyRef.jsx`	Inline link to methodology with disclaimer
FunFact	`components/FunFact.jsx`	Sunshine callout box for memorable comparisons
InsightTip	`components/InsightTip.jsx`	Reduction tip card with user data interpolated

7. State Management

All state is in-memory React state. No localStorage. No backend. Data is lost on page refresh — this is intentional and is disclosed to users ('Re-upload your file to analyze again').

7.1 useEcoData Hook (src/hooks/useEcoData.js)

Central state hook. Holds all parsed and computed data.

```
const {
  rawConversations,    // parsed JSON array
  processedData,       // tokenized + calculated per-conversation
  totals,              // { carbon_gco2e, water_liters, energy_kwh,
                       //   total_tokens, input_tokens, output_tokens,
                       //   conversation_count, message_count }
  dateRange,           // { earliest, latest }
  isLoaded,            // boolean
  isProcessing,        // boolean
  error,               // string | null
  uploadFile,          // (file: File) => void
  clearData,           // () => void
} = useEcoData()
```

7.2 useCoefficients Hook (src/hooks/useCoefficients.js)

Reads from src/data/coefficients.js. Exposes coefficients and version info. Makes future updates testable in isolation.

8. Data & Configuration Files

8.1 src/data/coefficients.js

THE most important file for future maintainers. Every coefficient has a comment with its source and date.

```
export const COEFFICIENTS = {
  version: '1.0',
  lastUpdated: 'March 2026',
  energy_per_output_token_wh: 0.000003,   // Luccioni et al. 2023
  energy_per_input_token_wh:  0.000001,   // Luccioni et al. 2023
  pue_multiplier:             1.2,         // industry avg estimate
  water_per_kwh_liters:       1.9,         // EESI 2024, direct cooling
  carbon_per_kwh_gco2e:       350,         // EPA eGRID 2023 US avg
  model_label:               'Claude Sonnet 4.6',
  uncertainty_range_pct:      50,          // honest ± range for all estimates
};
```

8.2 src/data/comparisons.js

Maps metric values to relatable comparisons. Each comparison has an emoji, label, divisor, and unit.

8.3 src/data/tips.js

Array of reduction tip objects. Each tip has: id, title, description, and a template string with {variable} placeholders that get filled with user-specific data at render time.

9. File & Folder Structure

`OpenH2O/`	Project root
`├── public/`	Static assets
`│  ├── logo.png`	OpenH2O / TAA logo
`│  └── favicon.ico`	
`├── src/`	Application source
`│  ├── main.jsx`	React entry point
`│  ├── App.jsx`	Router setup
`│  ├── index.css`	Tailwind + custom animations (copy from Susu)
`│  ├── components/`	Reusable UI components
`│  │  ├── Layout.jsx`	Sidebar + mobile nav shell
`│  │  ├── MetricCard.jsx`	Impact metric display
`│  │  ├── UploadZone.jsx`	File upload drag-drop
`│  │  ├── UsageChart.jsx`	Monthly bar chart
`│  │  ├── ConvoTable.jsx`	Conversation breakdown table
`│  │  ├── ComparisonBadge.jsx`	Real-world comparison callout
`│  │  ├── FunFact.jsx`	Sunshine fact box
`│  │  ├── InsightTip.jsx`	Reduction tip card

`│  │  ├── ProgressBar.jsx`	Copied from Susu unchanged
`│  │  └── MethodologyRef.jsx`	Inline methodology disclaimer link
`│  ├── pages/`	Route-level page components
`│  │  ├── Landing.jsx`	/ — Upload + hero
`│  │  ├── Processing.jsx`	/processing — Processing spinner
`│  │  ├── Dashboard.jsx`	/dashboard — Main impact view
`│  │  ├── Breakdown.jsx`	/breakdown — Conversation table
`│  │  ├── Insights.jsx`	/insights — Tips + patterns
`│  │  ├── Methodology.jsx`	/methodology — Full disclosure
`│  │  └── About.jsx`	/about — TAA + OpenH2O info
`│  ├── hooks/`	Custom React hooks
`│  │  ├── useEcoData.js`	Central data + processing state
`│  │  └── useCoefficients.js`	Coefficient access + versioning
`│  ├── data/`	Config and static data
`│  │  ├── coefficients.js`	⭐ All energy/water/carbon coefficients
`│  │  ├── comparisons.js`	Real-world comparison definitions
`│  │  └── tips.js`	Reduction tip templates
`│  └── utils/`	Pure utility functions
`│  ├── tokenizer.js`	gpt-tokenizer wrapper + helpers
`│  ├── calculator.js`	Core calculation engine
`│  └── formatters.js`	Number formatting, unit conversion
`├── METHODOLOGY.md`	⭐ Human-readable methodology doc for maintainers
`├── package.json`	
`├── vite.config.js`	
`└── tailwind.config.js`	

10. Dependencies

Package	Version	Purpose
`react`	`^19`	UI framework
`react-dom`	`^19`	DOM rendering
`react-router-dom`	`^7`	Client-side routing
`vite`	`^6`	Build tool
`tailwindcss`	`^4`	Utility CSS (matches Susu stack)
`recharts`	`latest`	Monthly usage bar chart
`gpt-tokenizer`	`latest`	Client-side BPE tokenization (GPT-4 compatible)
`lucide-react`	`^0.263`	Icons (matches Susu)

11. Privacy, Ethics & Disclosure

11.1 Privacy Architecture

- Zero server-side processing — all computation runs in the user's browser via JavaScript
- No API calls, no analytics, no telemetry, no cookies, no accounts
- Uploaded JSON file is never transmitted over the network
- Data is held in React state and garbage-collected when the user closes or refreshes the tab
- Privacy disclosure is shown on the landing page, the upload zone, and the about page

11.2 Estimation Honesty Requirements

OpenH2O must never misrepresent its estimates as precise measurements. The following disclosures must appear in the UI:

- Disclaimer on every metric: 'Estimated — see Methodology for details'
- Uncertainty range on dashboard: all figures shown as 'approximately X (±50%)'
- Model assumption disclosure: 'Calculated using Claude Sonnet 4.6 coefficients — your export does not include per-message model data'
- Tokenizer disclosure: 'Token counts use GPT-4 BPE encoding — within ~5% of Claude's actual tokenizer'
- Version + date on methodology page: coefficients version number and last-reviewed date

11.3 Open Source Requirements

- Repository: github.com/tech-awareness-association/OpenH2O (public)
- License: MIT
- METHODOLOGY.md in repo root: plain-English explanation of every calculation
- CONTRIBUTING.md: instructions for updating coefficients when new research is published
- Coefficient update PRs should include: new value, old value, source, date

12. Handoff & Maintenance Requirements

OpenH2O is built to be handed off to future TAA leadership. The following non-negotiable requirements ensure the tool remains accurate and maintainable after the original developer (Ronit) transitions out:

12.1 The Coefficient Update Protocol

When new energy/water/carbon research is published, a future maintainer must be able to update the tool with no knowledge of React or the codebase. The update path is:

- Open src/data/coefficients.js
- Update the relevant value(s) with the new figure
- Update the source comment with the new paper/report citation
- Increment coefficients_version by 0.1

- Update lastUpdated to current month/year
- Commit — the UI automatically picks up all changes

The coefficients file must remain self-contained and human-readable. No other file should hardcode any numerical environmental values.

12.2 Documentation Requirements

- Every function in calculator.js, tokenizer.js, and formatters.js must have a JSDoc comment explaining inputs, outputs, and units
- METHODOLOGY.md must be updated whenever coefficients.js changes
- README.md must include: project purpose, how to run locally, how to deploy, how to update coefficients

12.3 Sustainability of the Tool

- No paid API keys or services — tool must remain functional indefinitely at zero cost
- No dependencies on external data sources that could break (all coefficients are hardcoded from literature)
- Vite + React + static hosting = minimal maintenance burden

13. Claude Code Instructions

When using this spec with Claude Code, provide the following context prompt before pasting sections of this document:

Suggested Claude Code context prompt:

You are building OpenH2O, a React + Vite + Tailwind CSS web app for Tech Awareness Association (TAA). It allows Claude users to upload their exported conversations.json and see their AI environmental impact (carbon, water, energy). All processing is 100% client-side — no backend. The UI must exactly match the Susu app's boxy design language: border-4 border-navy on all cards, sharp corners (no border-radius), shadow-[8px_8px_0px_0px_#FAC206] on featured cards, bg-sunshine for primary CTAs, bg-navy sidebar, bg-cream (#F8F3F1) page background, font-black uppercase headings, Inter font. Color tokens: sunshine=#FAC206, sky=#16C0FF, coral=#FB4B5F, navy=#2C3E50, cream=#F8F3F1, ink=#181818, slate=#8B9B9C. The tech stack is identical to Susu (React Router, Tailwind v4). Refer to this spec document for all page layouts, component specs, data structures, and calculation logic.

Build order recommendation for Claude Code sessions:

- Session 1: Project scaffold, Tailwind config, Layout.jsx, index.css (copy animations from Susu)
- Session 2: src/data/coefficients.js, src/utils/tokenizer.js, src/utils/calculator.js, src/utils/formatters.js
- Session 3: useEcoData hook, UploadZone component, Landing page, Processing page
- Session 4: MetricCard, UsageChart, Dashboard page
- Session 5: ConvoTable, Breakdown page, search/sort functionality
- Session 6: InsightTip, ComparisonBadge, FunFact, Insights page
- Session 7: Methodology page, About page, final polish + README